

BINOMIAL THEOREM

1. The coefficient of x^{256} in the expansion of $(1-x)^{101}(x^2+x+1)^{100}$ is :

(1) $^{100}C_{16}$ (2) $^{100}C_{15}$
(3) $-^{100}C_{16}$ (4) $-^{100}C_{15}$

2. The number of rational terms in the binomial expansion of $\left(4^{\frac{1}{4}} + 5^{\frac{1}{6}}\right)^{120}$ is _____.

3. For the natural numbers m, n, if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \dots + a_{m+n}y^{m+n}$ and $a_1 = a_2 = 10$, then the value of (m + n) is equal to :

(1) 88 (2) 64 (3) 100 (4) 80

4. For $k \in \mathbb{N}$,

$$\text{let } \frac{1}{\alpha(\alpha+1)(\alpha+2)\dots(\alpha+20)} = \sum_{k=0}^{20} \frac{A_k}{\alpha+k},$$

where $\alpha > 0$. Then the value of

$$100 \left(\frac{A_{14} + A_{15}}{A_{13}} \right)^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

5. If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to _____.

6. The number of elements in the set $\{n \in \{1, 2, 3, \dots, 100\} \mid (11)^n > (10)^n + (9)^n\}$ is _____.

7. If b is very small as compared to the value of a, so that the cube and other higher powers of $\frac{b}{a}$

can be neglected in the identity

$$\frac{1}{a-b} + \frac{1}{a-2b} + \frac{1}{a-3b} + \dots + \frac{1}{a-nb} = \alpha n + \beta n^2 + \gamma n^3,$$

then the value of γ is :

(1) $\frac{a^2+b}{3a^3}$ (2) $\frac{a+b}{3a^2}$
(3) $\frac{b^2}{3a^3}$ (4) $\frac{a+b^2}{3a^3}$

8. The ratio of the coefficient of the middle term in the expansion of $(1+x)^{20}$ and the sum of the coefficients of two middle terms in expansion of $(1+x)^{19}$ is _____.

9. The term independent of 'x' in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$, where $x \neq 0, 1$ is equal to _____.

10. The sum of all those terms which are rational numbers in the expansion of $(2^{1/3} + 3^{1/4})^{12}$ is:

(1) 89 (2) 27 (3) 35 (4) 43

11. If the greatest value of the term independent of 'x' in the expansion of $\left(x \sin \alpha + a \frac{\cos \alpha}{x}\right)^{10}$ is $\frac{10!}{(5!)^2}$, then the value of 'a' is equal to :

(1) -1 (2) 1 (3) -2 (4) 2

12. The lowest integer which is greater than $\left(1 + \frac{1}{10^{100}}\right)^{10^{100}}$ is _____.

(1) 3 (2) 4 (3) 2 (4) 1

13. Let $n \in \mathbb{N}$ and $[x]$ denote the greatest integer less than or equal to x. If the sum of (n + 1) terms ${}^nC_0, 3 \cdot {}^nC_1, 5 \cdot {}^nC_2, 7 \cdot {}^nC_3, \dots$ is equal to

$$2^{100} \cdot 101, \text{ then } 2 \left[\frac{n-1}{2} \right] \text{ is equal to } \underline{\hspace{2cm}}.$$

14. If the co-efficient of x^7 and x^8 in the expansion of $\left(2 + \frac{x}{3}\right)^n$ are equal, then the value of n is equal to _____.

15. If the coefficients of x^7 in $\left(x^2 + \frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x - \frac{1}{bx^2}\right)^{11}$, $b \neq 0$, are equal, then the value of b is equal to:
 (1) 2 (2) -1 (3) 1 (4) -2
16. A possible value of 'x', for which the ninth term in the expansion of $\left\{3^{\log_3 \sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3 (5^{x-1}+1)}\right\}^{10}$ in the increasing powers of $3^{\left(-\frac{1}{8}\right)\log_3 (5^{x-1}+1)}$ is equal to 180, is :
 (1) 0 (2) -1 (3) 2 (4) 1
17. If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1+x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 \cdot {}^{20}C_r$ is equal to :
 (1) 420×2^{19} (2) 380×2^{19}
 (3) 380×2^{18} (4) 420×2^{18}
18. If ${}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + 15 \cdot {}^{15}P_{15} = {}^nP_r - s$, $0 \leq s \leq 1$, then ${}^{q+s}C_{r-s}$ is equal to _____.
19. Let $\binom{n}{k}$ denotes nC_k and $\left[\begin{matrix} n \\ k \end{matrix} \right] = \begin{cases} \binom{n}{k}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$
 If $A_k = \sum_{i=0}^9 \binom{9}{i} \left[\begin{matrix} 12 \\ 12-k+i \end{matrix} \right] + \sum_{i=0}^8 \binom{8}{i} \left[\begin{matrix} 13 \\ 13-k+i \end{matrix} \right]$
 and $A_4 - A_3 = 190$ p, then p is equal to :
20. If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots$, is equal to :
 (1) $x \left(\frac{1+x}{1-x} \right) + \log_e(1-x)$
 (2) $x \left(\frac{1-x}{1+x} \right) + \log_e(1-x)$
 (3) $\frac{1-x}{1+x} + \log_e(1-x)$
 (4) $\frac{1+x}{1-x} + \log_e(1-x)$
21. $\sum_{k=0}^{20} \left({}^{20}C_k \right)^2$ is equal to :
 (1) ${}^{40}C_{21}$ (2) ${}^{40}C_{19}$ (3) ${}^{40}C_{20}$ (4) ${}^{41}C_{20}$
22. $3 \times 7^{22} + 2 \times 10^{22} - 44$ when divided by 18 leaves the remainder _____.
23. If $\left(\frac{3^6}{4^4} \right)^k$ is the term, independent of x , in the binomial expansion of $\left(\frac{x}{4} - \frac{12}{x^2} \right)^{12}$, then k is equal to _____.
24. If the coefficient of a^7b^8 in the expansion of $(a+2b+4ab)^{10}$ is $K \cdot 2^{16}$, then K is equal to _____.
25. If the sum of the coefficients in the expansion of $(x+y)^n$ is 4096, then the greatest coefficient in the expansion is _____.
26. If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^nC_2 + {}^nC_3 + {}^nC_4 + \dots + {}^nC_n)$ is:
 (1) $\frac{n(n-1)(2n+1)}{6}$ (2) $\frac{n(n+1)(2n+1)}{6}$
 (3) $\frac{n(2n+1)(3n+1)}{6}$ (4) $\frac{n(n+1)^2(n+2)}{12}$
27. For integers n and r , let $\binom{n}{r} = \begin{cases} {}^nC_r, & \text{if } n \geq r \geq 0 \\ 0, & \text{otherwise} \end{cases}$
 The maximum value of k for which the sum $\sum_{i=0}^k \binom{10}{i} \binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i} \binom{13}{k+1-i}$ exists, is equal to _____.
28. The value of $-{}^{15}C_1 + 2 \cdot {}^{15}C_2 - 3 \cdot {}^{15}C_3 + \dots - 15 \cdot {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \dots + {}^{14}C_{11}$ is :
 (1) $2^{16} - 1$ (2) $2^{13} - 14$
 (3) 2^{14} (4) $2^{13} - 13$
29. If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is _____.

30. Let $m, n \in \mathbb{N}$ and $\gcd(2, n) = 1$. If

$$30 \binom{30}{0} + 29 \binom{30}{1} + \dots + 2 \binom{30}{28} + 1 \binom{30}{29} = n \cdot 2^m,$$

then $n + m$ is equal to

(Here $\binom{n}{k} = {}^nC_k$)

31. The maximum value of the term independent of

't' in the expansion of $\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t} \right)^{10}$

where $x \in (0, 1)$ is :

(1) $\frac{10!}{\sqrt{3}(5!)^2}$ (2) $\frac{2 \cdot 10!}{3\sqrt{3}(5!)^2}$

(3) $\frac{2 \cdot 10!}{3(5!)^2}$ (4) $\frac{10!}{3(5!)^2}$

32. Let n be a positive integer. Let

$$A = \sum_{k=0}^n (-1)^k n C_k \left[\left(\frac{1}{2} \right)^k + \left(\frac{3}{4} \right)^k + \left(\frac{7}{8} \right)^k + \left(\frac{15}{16} \right)^k + \left(\frac{31}{32} \right)^k \right]$$

If $63A = 1 - \frac{1}{2^{30}}$, then n is equal to _____.

33. If n is the number of irrational terms in the expansion of $(3^{1/4} + 5^{1/8})^{60}$, then $(n - 1)$ is divisible by :

(1) 26 (2) 30 (3) 8 (4) 7

34. Let $[x]$ denote greatest integer less than or equal

to x . If for $n \in \mathbb{N}$, $(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$, then

$$\sum_{j=0}^{\left[\frac{3n}{2} \right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2} \right]} a_{2j} + 1 \text{ is equal to :}$$

(1) 2 (2) 2^{n-1} (3) 1 (4) n

35. The value of $\sum_{r=0}^6 ({}^6C_r \cdot {}^6C_{6-r})$ is equal to :

(1) 1124 (2) 1324 (3) 1024 (4) 924

36. Let the coefficients of third, fourth and fifth

terms in the expansion of $\left(x + \frac{a}{x^2} \right)^n$, $x \neq 0$, be

in the ratio 12 : 8 : 3. Then the term independent of x in the expansion, is equal to _____.

37. Two dices are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

(1) $\frac{4}{9}$ (2) $\frac{17}{36}$ (3) $\frac{5}{12}$ (4) $\frac{1}{2}$

38. If the fourth term in the expansion of $(x + x^{\log_2 x})^7$ is 4480, then the value of x where $x \in \mathbb{N}$ is equal to :

(1) 2 (2) 4 (3) 3 (4) 1

39. The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is :

(1) $2 + \frac{2}{5}\sqrt{30}$ (2) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$

(3) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$ (4) $5 + \frac{2}{5}\sqrt{30}$

40. If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

41. Let $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$. then $a_1 + a_3 + a_5 + \dots + a_{37}$ is equal to

(1) $2^{20}(2^{20} - 21)$ (2) $2^{19}(2^{20} - 21)$

(3) $2^{19}(2^{20} + 21)$ (4) $2^{20}(2^{20} + 21)$

42. If $\sum_{r=1}^{10} r!(r^3 + 6r^2 + 2r + 5) = \alpha(11!)$, then the value of α is equal to _____.

43. Let $P(x)$ be a real polynomial of degree 3 which vanishes at $x = -3$. Let $P(x)$ have local minima at $x = 1$, local maxima at $x = -1$ and $\int_{-1}^1 P(x) dx = 18$, then the sum of all the coefficients of the polynomial $P(x)$ is equal to _____.
44. Let nC_r denote the binomial coefficient of x^r in the expansion of $(1+x)^n$.
If $\sum_{k=0}^{10} (2^2 + 3k) {}^nC_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$, $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is equal to _____.

SOLUTION

1. Official Ans. by NTA (2)

$$\begin{aligned}
 \text{Sol. } (1-x)^{100} \cdot (x^2+x+1)^{100} \cdot (1-x) \\
 &= ((1-x)(x^2+x+1))^{100} (1-x) \\
 &= (1^3 - x^3)^{100} (1-x) \\
 &= (1-x^3)^{100} (1-x) \\
 &= \underbrace{(1-x^3)^{100}}_{\text{No term of } x^{256}} - \underbrace{x(1-x^3)^{100}}_{\text{We find coefficient of } x^{255}} \\
 &\text{Required coefficient } (-1) \times (-^{100}C_{85}) \\
 &= {}^{100}C_{85} = {}^{100}C_{15}
 \end{aligned}$$

2. Official Ans. by NTA (21)

$$\begin{aligned}
 \text{Sol. } \left(4^{\frac{1}{4}} + 5^{\frac{1}{6}}\right)^{120} \\
 T_{r+1} = {}^{120}C_r (2^{1/2})^{120-r} (5)^{r/6} \\
 \text{for rational terms } r = 6\lambda \quad 0 \leq r \leq 120 \\
 \text{so total no of forms are 21.}
 \end{aligned}$$

3. Official Ans. by NTA (4)

$$\begin{aligned}
 \text{Sol. } (1-y)^m (1+y)^n \\
 \text{Coefficient of } y (a_1) = 1 \cdot {}^nC_1 + {}^mC_1 (-1) \\
 = n - m = 10 \quad \dots (1) \\
 \text{Coefficient of } y^2 (a_2) \\
 = 1 \cdot {}^nC_2 - {}^mC_1 \cdot {}^nC_1 + 1 \cdot {}^mC_2 = 10 \\
 = \frac{n(n-1)}{2} - m \cdot n + \frac{m(m-1)}{2} = 10 \\
 m^2 + n^2 - 2mn - (n+m) = 20 \\
 (n-m)^2 - (n+m) = 20 \\
 n+m = 80 \quad \dots (2) \\
 \text{By equation (1) \& (2)} \\
 m = 35, n = 45
 \end{aligned}$$

4. Official Ans. by NTA (9)

$$\begin{aligned}
 \text{Sol. } \frac{1}{\alpha(\alpha+1)\dots(\alpha+20)} &= \sum_{k=0}^{20} \frac{A_k}{\alpha+k} \\
 A_{14} &= \frac{1}{(-14)(-13)\dots(-1)(1)\dots(6)} = \frac{1}{14! \cdot 6!} \\
 A_{15} &= \frac{1}{(-15)(-14)\dots(-1)(1)\dots(5)} = \frac{1}{15! \cdot 5!} \\
 A_{13} &= \frac{1}{(-13)\dots(-1)(1)\dots(7)} = \frac{-1}{13! \cdot 7!} \\
 \frac{A_{14}}{A_{13}} &= \frac{1}{14! \cdot 6!} \times -13! \times 7! = \frac{-7}{14} = -\frac{1}{2} \\
 \frac{A_{15}}{A_{13}} &= -\frac{1}{15! \times 5!} \times -13! \times 7! = \frac{42}{15 \times 14} = \frac{1}{5} \\
 100 \left(\frac{A_{14}}{A_{13}} + \frac{A_{15}}{A_{13}} \right)^2 &= 100 \left(-\frac{1}{2} + \frac{1}{5} \right)^2 = 9
 \end{aligned}$$

5. Official Ans. by NTA (8)

$$\begin{aligned}
 \text{Sol. } \left(2x^r + \frac{1}{x^2}\right)^{10} \\
 \text{General term} = {}^{10}C_R (2x^2)^{10-R} x^{-2R} \\
 \Rightarrow 2^{10-R} {}^{10}C_R = 180 \quad \dots (1) \\
 \& (10-R)r - 2R = 0 \\
 r = \frac{2R}{10-R} \\
 r = \frac{2(R-10)}{10-R} + \frac{20}{10-R} \\
 \Rightarrow r = -2 + \frac{20}{10-R} \quad \dots (2) \\
 R = 8 \text{ or } 5 \text{ reject equation (1) not satisfied} \\
 \text{At } R = 8 \\
 2^{10-R} {}^{10}C_R = 180 \Rightarrow \boxed{r=8}
 \end{aligned}$$

6. Official Ans. by NTA (96)**Sol.** $11^n > 10^n + 9^n$

$$\begin{aligned} \Rightarrow 11^n - 9^n &> 10^n \\ \Rightarrow (10+1)^n - (10-1)^n &> 10^n \\ \Rightarrow \{ {}^nC_1 \cdot 10^{n-1} + {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} &> 10^n \\ \Rightarrow 2n \cdot 10^{n-1} + 2 \{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} &> 10^n \end{aligned}$$

.... (1)

For $n = 5$

$$10^5 + 2 \{ {}^5C_3 10^2 + {}^5C_5 \} > 10^5 \text{ (True)}$$

For $n = 6, 7, 8, \dots, 100$

$$2n10^{n-1} > 10^n$$

$$\Rightarrow 2n10^{n-1} + 2 \{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} > 10^n$$

$$\Rightarrow 11^n - 9^n > 10^n \text{ For } n = 5, 6, 7, \dots, 100$$

For $n = 4$, Inequality (1) is not satisfied $\Rightarrow$ Inequality does not hold good for

$$N = 1, 2, 3, 4$$

So, required number of elements

$$= 96$$

7. Official Ans. by NTA (3)**Sol.** $(a-b)^{-1} + (a-2b)^{-1} + \dots + (a-nb)^{-1}$

$$= \frac{1}{a} \sum_{r=1}^n \left(1 - \frac{rb}{a} \right)^{-1}$$

$$= \frac{1}{a} \sum_{r=1}^n \left\{ \left(1 + \frac{rb}{a} + \frac{r^2 b^2}{a^2} \right) + (\text{terms to be neglected}) \right\}$$

$$= \frac{1}{a} \left[n + \frac{n(n+1)}{2} \cdot \frac{b}{a} + \frac{n(n+1)(2n+1)}{6} \cdot \frac{b^2}{a^2} \right]$$

$$= \frac{1}{a} \left[n^3 \left(\frac{b^2}{3a^2} \right) + \dots \right]$$

$$\text{So } \gamma = \frac{b^2}{3a^3}$$

8. Official Ans. by NTA (1)**Sol.** Coeff. of middle term in $(1+x)^{20} = {}^{20}C_{10}$

& Sum of Coeff. of two middle terms in

$$(1+x)^{19} = {}^{19}C_9 + {}^{19}C_{10}$$

$$\text{So required ratio} = \frac{{}^{20}C_{10}}{{}^{19}C_9 + {}^{19}C_{10}} = \frac{{}^{20}C_{10}}{{}^{20}C_{10}} = 1$$

9. Official Ans. by NTA (210)

$$\text{Sol. } \left(\left(x^{1/3} + 1 \right) - \left(\frac{x^{1/2} + 1}{x^{1/2}} \right) \right)^{10}$$

$$= \left(x^{1/3} - \frac{1}{x^{1/2}} \right)^{10}$$

Now General Term

$$T_{r+1} = {}^{10}C_r \left(x^{1/3} \right)^{10-r} \cdot \left(-\frac{1}{x^{1/2}} \right)^r$$

For independent term

$$\frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow r = 4$$

$$\Rightarrow T_5 = {}^{10}C_4 = 210$$

10. Official Ans. by NTA (4)

$$\text{Sol. } T_{r+1} = {}^{12}C_r \left(2^{1/3} \right)^r \cdot \left(3^{1/4} \right)^{12-r}$$

 T_{r+1} will be rational numberwhen $r = 0, 3, 6, 9, 12$ & $r = 0, 4, 8, 12$

$$\Rightarrow r = 0, 12$$

$$T_1 + T_{13} = 1 \times 3^3 + 1 \times 2^4 \times 1$$

$$= 24 + 16 = 40$$

11. Official Ans. by NTA (4)

$$\text{Sol. } T_{r+1} = {}^{10}C_r (x \sin \alpha)^{10-r} \left(\frac{a \cos \alpha}{x} \right)^r$$

$$r = 0, 1, 2, \dots, 10$$

 T_{r+1} will be independent of x

$$\text{when } 10 - 2r = 0 \Rightarrow r = 5$$

$$T_6 = {}^{10}C_5 (x \sin \alpha)^5 \times \left(\frac{a \cos \alpha}{x} \right)^5$$

$$= {}^{10}C_5 \times a^5 \times \frac{1}{2^5} (\sin 2\alpha)^5$$

will be greatest when $\sin 2\alpha = 1$

$$\Rightarrow {}^{10}C_5 \frac{a^5}{2^5} = {}^{10}C_5 \Rightarrow a = 2$$

12. Official Ans. by NTA (1)

Sol. Let $P = \left(1 + \frac{1}{10^{100}}\right)^{10^{100}}$,
 Let $x = 10^{100}$
 $\Rightarrow P = \left(1 + \frac{1}{x}\right)^x$
 $\Rightarrow P = 1 + (x) \left(\frac{1}{x}\right) + \frac{(x)(x-1)}{2!} \cdot \frac{1}{x^2}$
 $+ \frac{(x)(x-1)(x-2)}{3!} \cdot \frac{1}{x^3} + \dots$

(upto $10^{100} + 1$ terms)

$$\Rightarrow P = 1 + 1 + \left(\frac{1}{2} - \frac{1}{2x^2}\right) + \left(\frac{1}{3} - \dots\right) + \dots \text{so on}$$

$$\Rightarrow P = 2 + \left(\text{Positive value less than } \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots\right)$$

Also $e = 1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

$$\Rightarrow \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = e - 2$$

$$\Rightarrow P = 2 + (\text{positive value less than } e - 2)$$

$$\Rightarrow P \in (2, 3)$$

$$\Rightarrow \text{least integer value of } P \text{ is } 3$$

13. Official Ans. by NTA (98)

Sol. $1 \cdot {}^nC_0 + 3 \cdot {}^nC_1 + 5 \cdot {}^nC_2 + \dots + (2n+1) \cdot {}^nC_n$

$$T_r = (2r+1) {}^nC_r$$

$$S = \sum T_r$$

$$S = \sum (2r+1) {}^nC_r = \sum 2r {}^nC_r + \sum {}^nC_r$$

$$S = 2(n \cdot 2^{n-1}) + 2^n = 2^n(n+1)$$

$$2^n(n+1) = 2^{100} \cdot 101 \Rightarrow n = 100$$

$$2 \left[\frac{n-1}{2} \right] = 2 \left[\frac{99}{2} \right] = 98$$

14. Official Ans. by NTA (55)

Sol. ${}^nC_7 2^{n-7} \frac{1}{3^7} = {}^nC_8 2^{n-8} \frac{1}{3^8}$

$$\Rightarrow n-7 = 48 \Rightarrow n = 55$$

15. Official Ans. by NTA (3)

Sol. Coefficient of x^7 in $\left(x^2 + \frac{1}{bx}\right)^{11}$

$${}^{11}C_r (x^2)^{11-r} \cdot \left(\frac{1}{bx}\right)^r$$

$${}^{11}C_r x^{22-3r} \cdot \frac{1}{b^r}$$

$$22-3r = 7$$

$$r = 5$$

$$\therefore {}^{11}C_5 \cdot \frac{1}{b^5} \cdot x^7$$

Coefficient of x^{-7} in $\left(x - \frac{b}{bx^2}\right)^{11}$

$${}^{11}C_r (x)^{11-r} \cdot \left(-\frac{1}{bx^2}\right)^r$$

$${}^{11}C_r x^{11-3r} \cdot \frac{(-1)^r}{b^r}$$

$$11-3r = -7 \therefore r = 6$$

$${}^{11}C_6 \cdot \frac{1}{b^6} x^{-7}$$

$${}^{11}C_5 \cdot \frac{1}{b^5} = {}^{11}C_6 \cdot \frac{1}{b^6}$$

$$\text{Since } b \neq 0 \therefore b = 1$$

16. Official Ans. by NTA (4)

Sol. ${}^{10}C_8 (25^{(x-1)} + 7) \times (5^{(x-1)} + 1)^{-1} = 180$

$$\Rightarrow \frac{25^{x-1} + 7}{5^{(x-1)} + 1} = 4$$

$$\Rightarrow \frac{t^2 + 7}{t + 1} = 4;$$

$$\Rightarrow t = 1, 3 = 5^{x-1}$$

$$\Rightarrow x - 1 = 0 \text{ (one of the possible value).}$$

$$\Rightarrow x = 1$$

17. Official Ans. by NTA (4)

$$\begin{aligned}
 \text{Sol. } & \sum_{r=0}^{20} r^2 \cdot {}^{20}C_r \\
 & \sum (4(r-1) + r) \cdot {}^{20}C_r \\
 & \sum r(r-1) \cdot \frac{20 \times 19}{r(r-1)} \cdot {}^{18}C_{r-2} + r \cdot \frac{20}{r} \cdot \sum {}^{19}C_{r-1} \\
 & \Rightarrow 20 \times 19 \cdot 2^{18} + 20 \cdot 2^{19} \\
 & \Rightarrow 420 \times 2^{18}
 \end{aligned}$$

18. Official Ans. by NTA (136)

$$\begin{aligned}
 \text{Sol. } & {}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + 15 \cdot {}^{15}P_{15} \\
 & = 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + 15 \times 15! \\
 & = \sum_{r=1}^{15} (r+1-1)r! \\
 & = \sum_{r=1}^{15} (r+1)! - (r)! \\
 & = 16! - 1 \\
 & = {}^{16}P_{16} - 1 \\
 & \Rightarrow q = r = 16, s = 1 \\
 & {}^{q+s}C_{r-s} = {}^{17}C_{15} = 136
 \end{aligned}$$

19. Official Ans. by NTA (49)

$$\begin{aligned}
 \text{Sol. } & A_k = \sum_{i=0}^9 {}^9C_i \cdot {}^{12}C_{k-i} + \sum_{i=0}^8 {}^8C_i \cdot {}^{13}C_{k-i} \\
 & A_k = {}^{21}C_k + {}^{21}C_k = 2 \cdot {}^{21}C_k \\
 & A_4 - A_3 = 2({}^{21}C_4 - {}^{21}C_3) = 2(5985 - 1330) \\
 & 190p = 2(5985 - 1330) \Rightarrow p = 49.
 \end{aligned}$$

20. Official Ans. by NTA (1)

$$\begin{aligned}
 \text{Sol. } & \text{Let } t = \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots \infty \\
 & = \left(2 - \frac{1}{2}\right)x^2 + \left(2 - \frac{1}{3}\right)x^3 + \left(2 - \frac{1}{4}\right)x^4 + \dots \infty \\
 & = 2(x^2 + x^3 + x^4 + \dots \infty) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty\right) \\
 & = \frac{2x^2}{1-x} - (\ln(1-x) - x) \\
 & \Rightarrow t = \frac{2x^2}{1-x} + x - \ln(1-x) \\
 & \Rightarrow t = \frac{x(1+x)}{1-x} - \ln(1-x)
 \end{aligned}$$

21. Official Ans. by NTA (3)

$$\begin{aligned}
 \text{Sol. } & \sum_{k=0}^{20} {}^{20}C_k \cdot {}^{20}C_{20-k} \\
 & \text{sum of suffix is const. so summation will be } {}^{40}C_{20}
 \end{aligned}$$

22. Official Ans. by NTA (15)

$$\begin{aligned}
 \text{Sol. } & 3(1+6)^{22} + 2 \cdot (1+9)^{22} - 44 = (3+2-44) = 18 \cdot I \\
 & = -39 + 18 \cdot I \\
 & = (54 - 39) + 18(I - 3) \\
 & = 15 + 18 I_1 \\
 & \Rightarrow \text{Remainder} = 15.
 \end{aligned}$$

23. Official Ans. by NTA (55)

$$\begin{aligned}
 \text{Sol. } & \left(\frac{x}{4} - \frac{12}{x^2}\right)^{12} \\
 & T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{x}{4}\right)^{12-r} \left(\frac{12}{x^2}\right)^r \\
 & T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{1}{4}\right)^{12-r} (12)^r \cdot (x)^{12-3r} \\
 & \text{Term independent of } x \Rightarrow 12 - 3r = 0 \Rightarrow r = 4 \\
 & T_5 = (-1)^4 \cdot {}^{12}C_4 \left(\frac{1}{4}\right)^8 (12)^4 = \frac{3^6}{4^4} \cdot k \\
 & \Rightarrow k = 55
 \end{aligned}$$

24. Official Ans. by NTA (315)

Sol. $\frac{10!}{\alpha!\beta!\gamma!} a^\alpha (2b)^\beta (4ab)^\gamma$

$$\frac{10!}{\alpha!\beta!\gamma!} a^{\alpha+\gamma} b^{\beta+\gamma} 2^\beta 4^\gamma$$

$$\alpha + \beta + \gamma = 10 \quad \dots(1)$$

$$\alpha + \gamma = 7 \quad \dots(2)$$

$$\beta + \gamma = 8 \quad \dots(3)$$

$$(2) + (3) - (1) \Rightarrow \gamma = 5$$

$$\alpha = 2$$

$$\beta = 3$$

$$\text{so coefficients} = \frac{10!}{2!3!5!} 2^3 \cdot 2^{10}$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{2 \times 3 \times 2 \times 5!} \times 2^{13}$$

$$= 315 \times 2^{16} \Rightarrow k = 315$$

25. Official Ans. by NTA (924)

Sol. $(x + y)^n \Rightarrow 2^n = 4096 \quad 2^{10} = 1024 \times 2$

$$\Rightarrow 2^n = 2^{12} \quad 2^{11} = 2048$$

$$n = 12 \quad 2^{12} = 4096$$

$${}^{12}C_6 = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= 11 \times 3 \times 4 \times 7$$

$$= 924$$

26. Official Ans. by NTA (2)

Sol. ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$

$${}^{n+1}C_2 + 2({}^3C_3 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$$

$$\left\{ \text{use } {}^nC_{r+1} + {}^nC_r = {}^{n+1}C_r \right\}$$

$$= {}^{n+1}C_2 + 2({}^4C_3 + {}^4C_2 + {}^5C_3 + \dots + {}^nC_2)$$

$$= {}^{n+1}C_2 + 2({}^5C_3 + {}^5C_2 + \dots + {}^nC_2)$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$= {}^{n+1}C_2 + 2({}^nC_3 + {}^nC_2)$$

$$= {}^{n+1}C_2 + 2 \cdot {}^{n+1}C_3$$

$$= \frac{(n+1)n}{2} + 2 \cdot \frac{(n+1)(n)(n-1)}{2 \cdot 3}$$

$$= \frac{n(n+1)(2n+1)}{6}$$

27. Official Ans. by NTA (BONUS)

Sol. Bonus

$$\sum_{i=0}^k \binom{10}{i} \binom{15}{k-i} + \sum_{i=0}^{k+1} \binom{12}{i} \binom{13}{k+1-i}$$

$${}^{25}C_k + {}^{25}C_{k+1}$$

$${}^{26}C_{k+1}$$

as nC_r is defined for all values of n as well as r

so ${}^{26}C_{k+1}$ always exists

Now k is unbounded so maximum value is not defined.

28. Official Ans. by NTA (2)

Sol. $(-{}^{15}C_1 + 2 \cdot {}^{15}C_2 - 3 \cdot {}^{15}C_3 + \dots - 15 \cdot {}^{15}C_{15})$

$$+ ({}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11})$$

$$= \sum_{r=1}^{15} (-1)^r \cdot r \cdot {}^{15}C_r + ({}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} + {}^{14}C_{13}) - {}^{14}C_3$$

$$= \sum_{r=1}^{15} (-1)^r \cdot 15 \cdot {}^{14}C_{r-1} + 2^{13} - 14$$

$$= 15(-{}^{14}C_0 + {}^{14}C_1 - \dots - {}^{14}C_{14}) + 2^{13} - 14$$

29. Official Ans. by NTA (1)**Sol.** $x = 4k + 3$

$$\therefore (2020 + x)^{2022} = (2020 + 4k + 3)^{2022}$$

$$= (4(505 + k) + 3)^{2022}$$

$$= (4\lambda + 3)^{2022} = (16\lambda^2 + 24\lambda + 9)^{1011}$$

$$= (8(2\lambda^2 + 3\lambda + 1) + 1)^{1011}$$

$$= (8p + 1)^{1011}$$

$$\therefore \text{Remainder when divided by } 8 = 1$$

$$= 2^{13} - 14$$

30. Official Ans. by NTA (45)**Sol.** $30({}^{30}C_0) + 29({}^{30}C_1) + \dots + 2({}^{30}C_{28}) + 1({}^{30}C_{29})$

$$= 30({}^{30}C_{30}) + 29({}^{30}C_{29}) + \dots + 2({}^{30}C_2) + 1({}^{30}C_1)$$

$$= \sum_{r=1}^{30} r({}^{30}C_r)$$

$$= \sum_{r=1}^{30} r \left(\frac{30}{r} \right) ({}^{29}C_{r-1})$$

$$= 30 \sum_{r=1}^{30} {}^{29}C_{r-1}$$

$$= 30({}^{29}C_0 + {}^{29}C_1 + {}^{29}C_2 + \dots + {}^{29}C_{29})$$

$$= 30(2^{29}) = 15(2)^{30} = n(2)^m$$

$$\therefore n = 15, m = 30$$

31. Official Ans. by NTA (2)**Sol.** Term independent of t will be the middle term due to exact same magnitude but opposite sign powers of t in the binomial expression given

$$\text{so } T_6 = {}^{10}C_5 (tx^2 5)^5 \left(\frac{(1-x)^{\frac{1}{10}}}{t} \right)^5$$

$$T_6 = f(x) = {}^{10}C_5 (x\sqrt{1-x}); \text{ for maximum}$$

$$f'(x) = 0 \Rightarrow x = \frac{2}{3} \text{ \& } f'' \left(\frac{2}{3} \right) < 0$$

$$\text{so } f(x)_{\max.} = {}^{10}C_5 \left(\frac{2}{3} \right) \cdot \frac{1}{\sqrt{3}}$$

32. Official Ans. by NTA (6)

$$\text{Sol. } A = \sum_{k=0}^n {}^nC_k \left[\left(-\frac{1}{2} \right)^k + \left(-\frac{3}{4} \right)^k + \left(-\frac{7}{8} \right)^k + \left(-\frac{15}{16} \right)^k + \left(-\frac{31}{32} \right)^k \right]$$

$$A = \left(1 - \frac{1}{2} \right)^n + \left(1 - \frac{3}{4} \right)^n + \left(1 - \frac{7}{8} \right)^n + \left(1 - \frac{15}{16} \right)^n + \left(1 - \frac{31}{32} \right)^n$$

$$A = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{8^n} + \frac{1}{16^n} + \frac{1}{32^n}$$

$$A = \frac{1}{2^n} \left(\frac{1 - \left(\frac{1}{2^n} \right)^5}{1 - \frac{1}{2^n}} \right) \Rightarrow A = \frac{\left(1 - \frac{1}{2^{5n}} \right)}{(2^n - 1)}$$

$$(2^n - 1)A = 1 - \frac{1}{2^{5n}}, \text{ Given } 63A = 1 - \frac{1}{2^{30}}$$

$$\text{Clearly } 5n = 30$$

$$n = 6$$

33. Official Ans. by NTA (1)**Sol.** $(3^{1/4} + 5^{1/8})^{60}$

$${}^{60}C_r (3^{1/4})^{60-r} \cdot (5^{1/8})^r$$

$${}^{60}C_r (3)^{\frac{60-r}{4}} \cdot 5^{\frac{r}{8}}$$

For rational terms.

$$\frac{r}{8} = k; \quad 0 \leq r \leq 60$$

$$0 \leq 8k \leq 60$$

$$0 \leq k \leq \frac{60}{8}$$

$$0 \leq k \leq 7.5$$

$$k = 0, 1, 2, 3, 4, 5, 6, 7$$

$$\frac{60-8k}{4} \text{ is always divisible by 4 for all value of } k.$$

$$\text{Total rational terms} = 8$$

$$\text{Total terms} = 61$$

$$\text{irrational terms} = 53$$

$$n - 1 = 53 - 1 = 52$$

$$52 \text{ is divisible by } 26.$$

34. Official Ans. by NTA (3)

Sol. $(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$

$$(1 - x + x^3)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{3n} x^{3n}$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} = \text{Sum of } a_0 + a_2 + a_4 + \dots$$

$$\sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = \text{Sum of } a_1 + a_3 + a_5 + \dots$$

put $x = 1$

$$1 = a_0 + a_1 + a_2 + a_3 + \dots + a_{3n} \quad \dots (A)$$

Put $x = -1$

$$1 = a_0 - a_1 + a_2 - a_3 + \dots + (-1)^{3n} a_{3n} \quad \dots (B)$$

Solving (A) and (B)

$$a_0 + a_2 + a_4 + \dots = 1$$

$$a_1 + a_3 + a_5 + \dots = 0$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = 1$$

35. Official Ans. by NTA (4)

Sol. $\sum_{r=0}^6 {}^6C_r \cdot {}^6C_{6-r}$

$$= {}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0$$

Now,

$$(1+x)^6 (1+x)^6$$

$$= ({}^6C_0 + {}^6C_1 x + {}^6C_2 x^2 + \dots + {}^6C_6 x^6)$$

$$({}^6C_0 + {}^6C_1 x + {}^6C_2 x^2 + \dots + {}^6C_6 x^6)$$

Comparing coefficient of x^6 both sides

$${}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0 = {}^{12}C_6$$

$$= 924$$

Ans.(4)

36. Official Ans. by NTA (4)

Sol. $T_{r+1} = {}^nC_r (x)^{n-r} \left(\frac{a}{x^2}\right)^r$

$$= {}^nC_r a^r x^{n-3r}$$

$${}^nC_2 a^2 : {}^nC_3 a^3 : {}^nC_4 a^4 = 12 : 8 : 3$$

After solving

$$n = 6, a = \frac{1}{2}$$

For term independent of 'x' $\Rightarrow n = 3r$

$$r = 2$$

$\therefore$ Coefficient is ${}^6C_2 \left(\frac{1}{2}\right)^2 = \frac{15}{4}$

Nearest integer is 4.

37. Official Ans. by NTA (2)

Sol. $n(E) = 5 + 4 + 4 + 3 + 1 = 17$

So, $P(E) = \frac{17}{36}$

38. Official Ans. by NTA (1)

Sol. ${}^7C_3 x^4 \cdot x^{(3 \log_2^x)} = 4480$

$$\Rightarrow x^{(4+3 \log_2^x)} = 2^7$$

$$\Rightarrow (4+3t)t = 7; t = \log_2^x$$

$$\Rightarrow t = 1, \frac{-7}{3} \Rightarrow x = 2$$

39. Official Ans. by NTA (1)

Sol. $y = 4 + \frac{1}{\left(5 + \frac{1}{y}\right)}$

$$y - 4 = \frac{y}{(5y + 1)}$$

$$5y^2 - 20y - 4 = 0$$

$$y = \frac{20 + \sqrt{480}}{10}$$

$$y = \frac{20 - \sqrt{480}}{10} \rightarrow \text{rejected}$$

$$y = 2 + \sqrt{\frac{480}{100}}$$

Correct with Option (A)

40. Official Ans. by NTA (4)

$$\begin{aligned}
 \text{Sol. } (2023 - 2)^{3762} &= 2023k_1 + 2^{3762} \\
 &= 17k_2 + 2^{3762} \text{ (as } 2023 = 17 \times 17 \times 9) \\
 &= 17k_2 + 4 \times 16^{940} \\
 &= 17k_2 + 4 \times (17 - 1)^{940} \\
 &= 17k_2 + 4(17k_3 + 1) \\
 &= 17k + 4 \Rightarrow \text{remainder} = 4
 \end{aligned}$$

41. Official Ans. by NTA (2)

$$\text{Sol. } (1 + x + 2x^2)^{20} = a_0 + a_1x + \dots + a_{40}x^{40}$$

$$\text{put } x = 1, -1$$

$$\Rightarrow a_0 + a_1 + a_2 + \dots + a_{40} = 2^{20}$$

$$a_0 - a_1 + a_2 - \dots + a_{40} = 2^{20}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{39} = \frac{4^{20} - 2^{20}}{2}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{39} - 2^{19} - a_{39}$$

$$\text{here } a_{39} = \frac{20!(2)^{19} \times 1}{19!} = 20 \times 2^{19}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{19}(2^{20} - 1 - 20)$$

$$= 2^{19}(2^{20} - 21)$$

42. Official Ans. by NTA (160)

$$\begin{aligned}
 \text{Sol. } \sum_{r=1}^{10} r! \{ (r+1)(r+2)(r+3) - 9(r+1) + 8 \} \\
 &= \sum_{r=1}^{10} [\{ (r+3)! - (r+1)! \} - 8 \{ (r+1)! - r! \}] \\
 &= (13! + 12! - 2! - 3!) - 8(11! - 1) \\
 &= (12 \cdot 13 + 12 - 8) \cdot 11! - 8 + 8 \\
 &= (160)(11!)
 \end{aligned}$$

$$\text{Hence } \alpha = 160$$

43. Official Ans. by NTA (8)

$$\text{Sol. Let } p'(x) = a(x-1)(x+1) = a(x^2 - 1)$$

$$p(x) = a \int (x^2 - 1) dx + c$$

$$= a \left(\frac{x^3}{3} - x \right) + c$$

$$\text{Now } p(-3) = 0$$

$$\Rightarrow a \left(-\frac{27}{3} + 3 \right) + c = 0$$

$$\Rightarrow -6a + c = 0 \quad \dots(1)$$

$$\text{Now } \int_{-1}^1 \left(a \left(\frac{x^3}{3} - x \right) + c \right) dx = 18$$

$$= 2c = 18 \Rightarrow c = 9 \quad \dots(2)$$

$$\Rightarrow \text{from (1) \& (2)} \Rightarrow -6a + 9 = 0 \Rightarrow a = \frac{3}{2}$$

$$\Rightarrow p(x) = \frac{3}{2} \left(\frac{x^3}{3} - x \right) + 9$$

sum of coefficient

$$= \frac{1}{2} - \frac{3}{2} + 9$$

$$= 8$$

44. Official Ans. by NTA (19)

Allen Answer (Bonus)

Sol. BONUS

Instead of nC_k it must be ${}^{10}C_k$ i.e.

$$\sum_{k=0}^{10} (2^2 + 3k) {}^{10}C_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$$

$$\text{LHS} = 4 \sum_{k=0}^{10} {}^{10}C_k + 3 \sum_{k=0}^{10} k \cdot \frac{10}{k} \cdot {}^9C_{k-1}$$

$$= 4 \cdot 2^{10} + 3 \cdot 10 \cdot 2^9$$

$$= 19 \cdot 2^{10} = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$$

$$\Rightarrow \alpha = 0, \beta = 19 \Rightarrow \alpha + \beta = 19$$